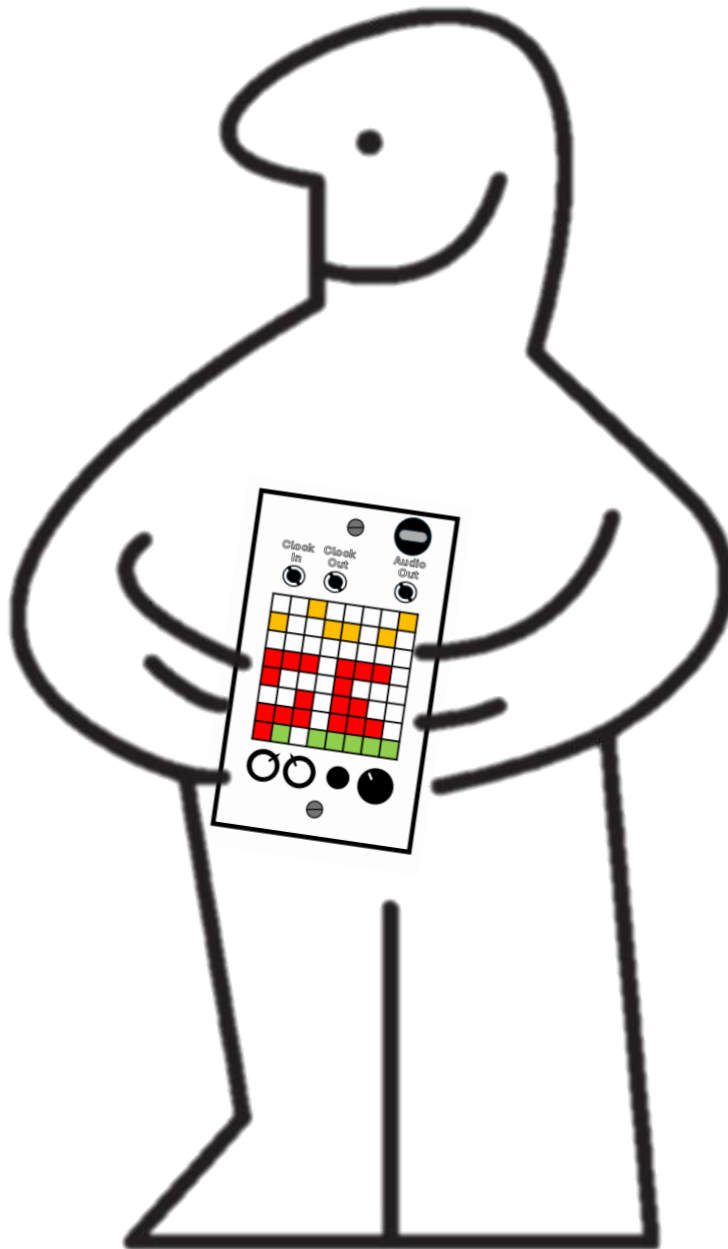


Getting to know your Prioritron



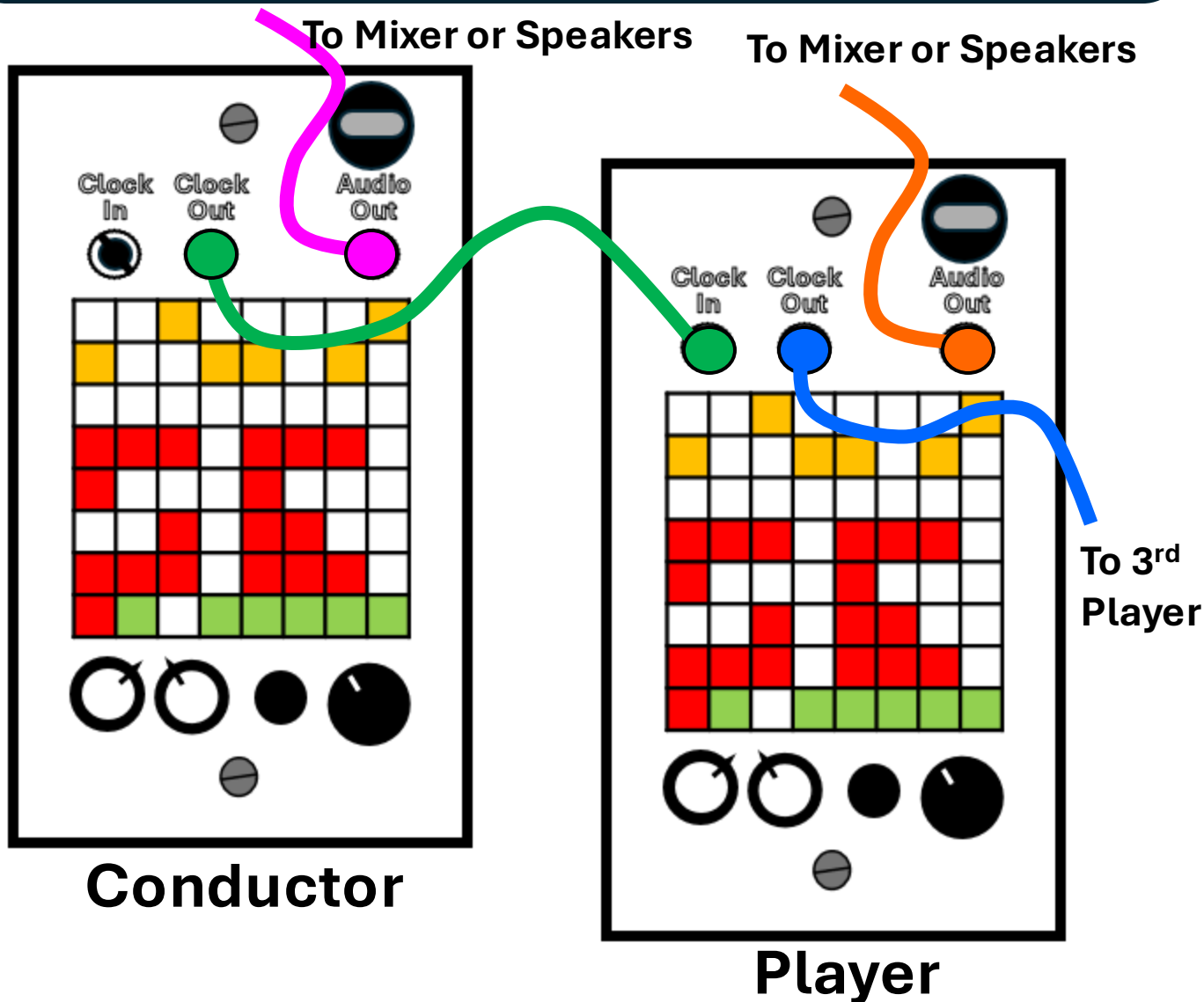
The Prioritron is a synthesiser and sequencer, created for a Creativity week activity at Priory School, Lewes in July 2025

How to connect two Prioritrons together to play in time

When the Prioritron is first turned on it looks for a clock signal on its “Clock in”. If it does not find one then it works independently, going into “**Conductor**” mode. This is indicated by a green square after the random multicoloured screen. In this mode you can change the tempo using the timing mode. It will also produce a clock signal on the Clock Out.

To get two Prioritrons to play in time,

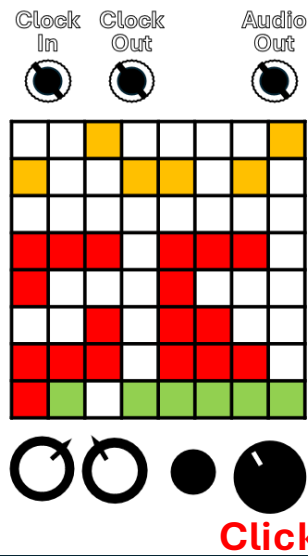
- 1) Turn on the first prioritron (that you want to use as **The Conductor**). This will show a green screen when it turns on.
- 2) Use a patch cable to connect the Clock Out from the **Conductor** to the Clock In for the **Player**
- 3) Turn on the **Player**. You should see a Pink Square. If you don't then check your connections.
- 4) You can add a third, or fourth Prioritron in the same way. Linking the Clock out from the first player to the Clock in of the next.



Menu Text: **Se**

Mode: **Sequencer**

In this mode you edit the notes in the current sequence.
The notes are in the same scale (which can be set on the scale menu). If you scroll off the top or the bottom of the screen then the note will switch off.
If you continue to scroll then the note will go into the octave above or below, shown by a change in colour.



Clock In Clock Out Audio Out

Scroll >> (vertical arrow)

Click ! (diagonal arrow)

Scroll >> (horizontal arrow)

Filter Envelope
Controls the amount of filter applied to the start of the note. Turn to far right to turn off filter.

Filter Cutoff
Controls the frequency of the filter

Scroll >> to change the step you are editing

Click ! to toggle between scrolling the step and scrolling the note

Long Click ...! Return to main menu

Click ! Save Current Sequence and switch to next sequence (indicated by colour)

Long Click ...! Save Sequence

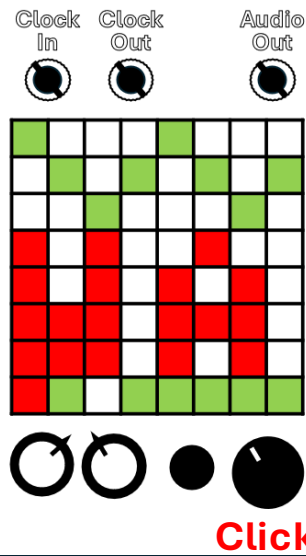
Hold and Scroll !>> If you hold this button and scroll then you will copy the current sequence into the sequence colour you are on when you release.

Menu Text: **WA**

Mode: **Waveforms**

In this mode you can switch between the waveforms.

Sine and Triangle waves can be very quiet for low pitched sounds. For basslines try Saw and Square waves.



Filter Cutoff
Controls the frequency of the filter

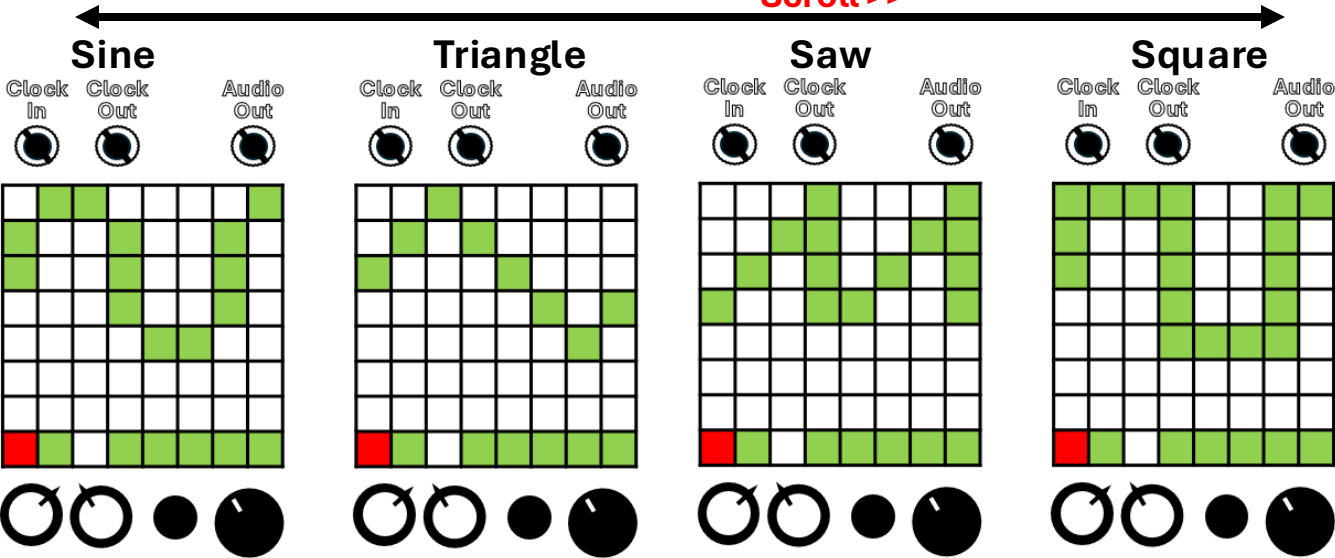
Filter Envelope
Controls the amount of filter applied to the start of the note. Turn to far right to turn off filter

Scroll >> to change the waveform

Long Click ...! Return to main menu

Click ! Change sequence

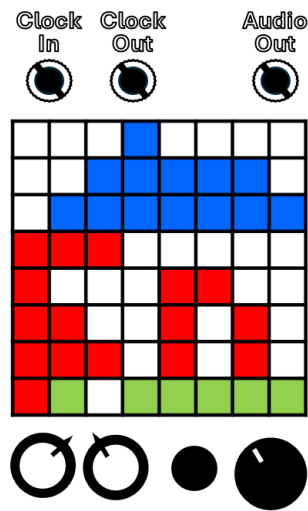
Scroll >>



Menu Text: En

Mode: Envelope

In this mode you modify the Attack, Decay, Sustain and Release (ADSR) of the note . The Attack is the time taken for the note to reach the maximum volume. The Decay is the time that the note takes to fall to a lower volume. The Sustain is the time the note holds for and the Release is the time it takes for the volume to reduce to zero. Short Attack times emphasise rhythm. Long sustain and decay times can create interesting harmonics when notes overlap.



Filter Cutoff
Controls the frequency of the filter

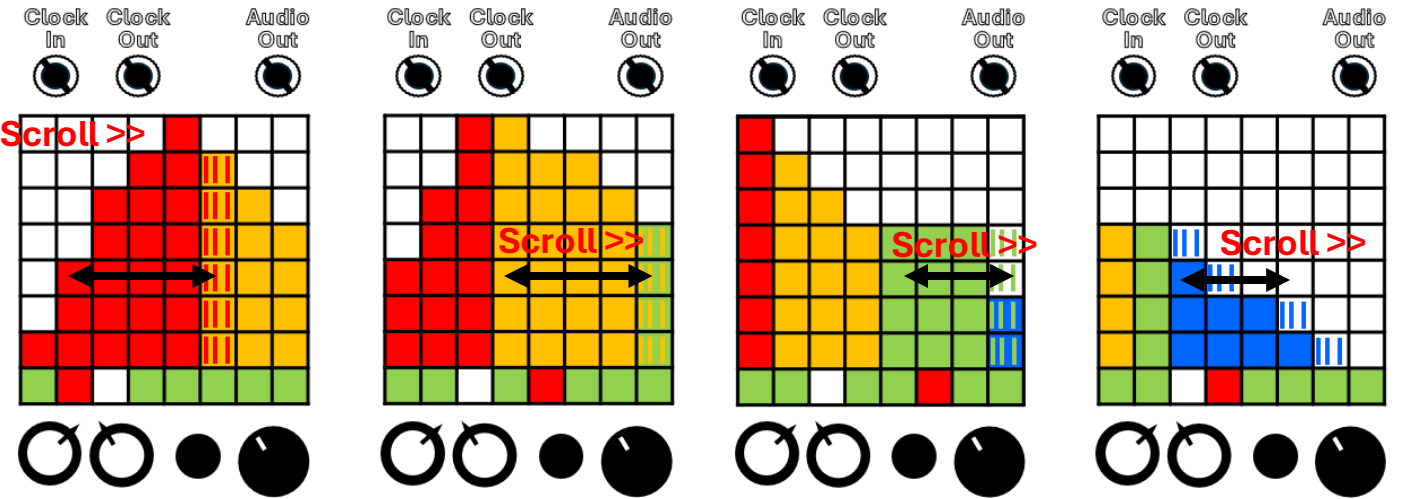
Filter Envelope
Controls the amount of filter applied to the start of the note. Turn to far right to turn off filter.

Click ! Change sequence

Click ! Toggle between selecting scrolling between parts of the envelope. Cursor is Red square.

Long Click ...! Return to main menu

- Attack (Red)**
This controls how quickly the note comes on
- Decay (Yellow)**
This part of the envelope controls the loudest part of the note
- Sustain (Green)**
This controls how long the note holds
- Release (Blue)**
This controls how the note dies away

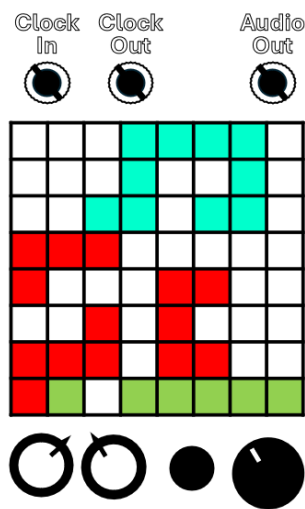


Menu Text: **Sc**

Mode: **Scale**

The Scale mode allows you to switch between major and minor scales and allows you to select a different tonic note.

The tonic note will be the first note from the bottom on the sequencer. For example, if you have selected a B major scale, the first note on the sequencer will be B, followed by **B – C# – D# – E – F# – G# – A#**.



Click!

Click ! Change sequence

Filter Cutoff
Controls the frequency of the filter

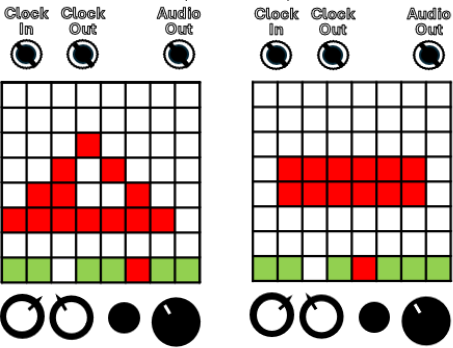
Filter Envelope
Controls the amount of filter applied to the start of the note.
Turn to far right to turn off filter.

Scroll >> to select major or minor scale and then tonic note.

Click ! Changes are only applied when you click. This is indicated by a colour change Red to Green

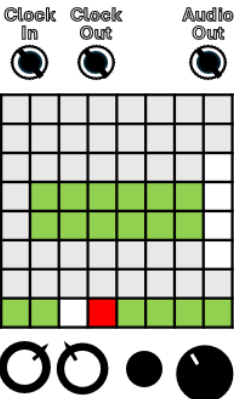
Long Click ...! Select Tonic Note

Scroll >> to see options

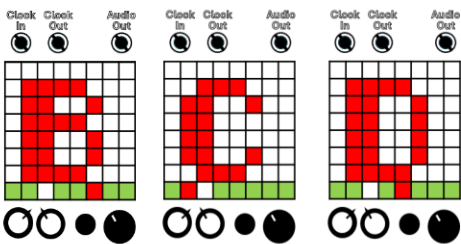


Click !
to select change

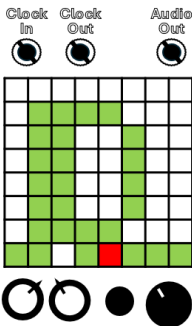
Long Click ...!
Select Tonic Note



Scroll >> to see options



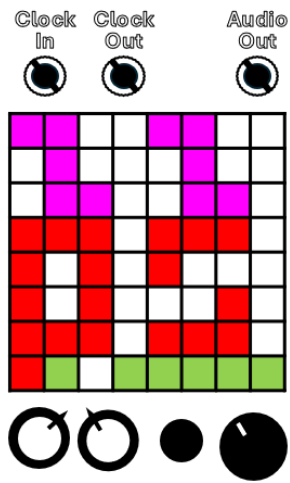
Click !
to select change



Menu Text: 2 OS

Mode: 2 Oscillator (Polyphonic)

In this mode the notes can be made to sound richer and more complex by playing a second note at the same time. This second oscillator can be programmed with an interval of 0 semi tones, 7 semi tones (a perfect 5th), 12 notes a whole octave or just a set frequency difference to give a detuned sound. As the Prioritron has 4 oscillators in total, in this mode it is possible to play two notes at the same time (when envelopes overlap)



Click!

Click ! Change sequence

Filter Cutoff
Controls the frequency of the filter

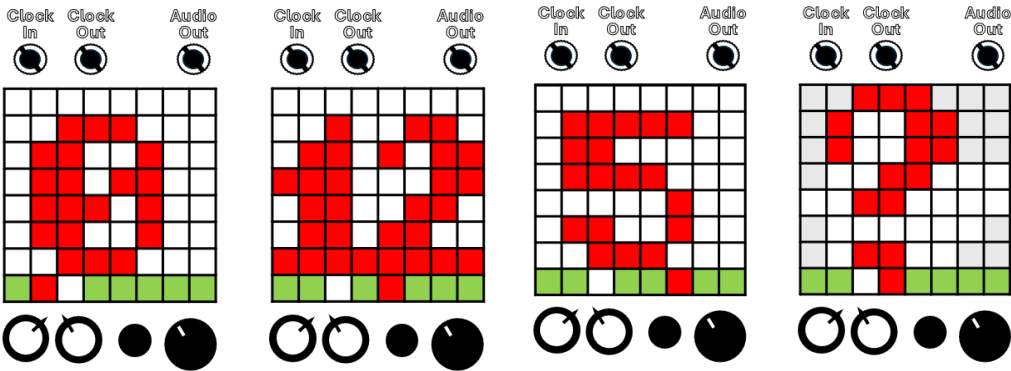
Filter Envelope
Controls the amount of filter applied to the start of the note. Turn to far right to turn off filter.

Scroll >> to select interval between oscillators

Click ! Changes are only applied when you click. This is indicated by a colour change Red to Green

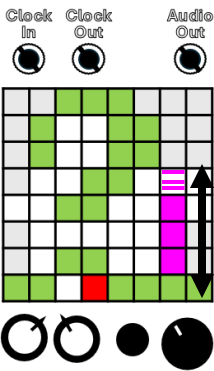
Long Click ...! Return to main menu

Scroll >> to see options



Click !
To select change (goes green)

Scroll >> to change offset by arbitrary frequency in this ? Detuned mode

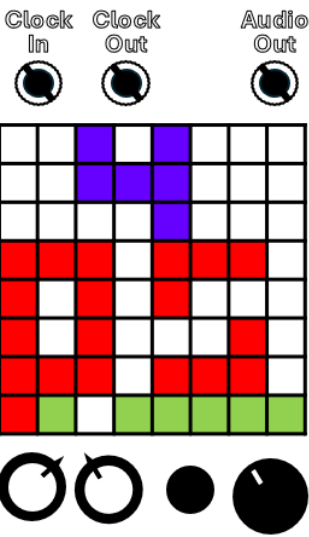


Menu Text: 4 OS

Mode: 4 Oscillator (Drone)

This mode is designed to make drones and chords. The attack, decay and sustain of the envelopes are made very long. There are 4 oscillators so four notes can be played simultaneously. If a 5th note is selected in the sequence then the first note stops as the next one begins.

In this mode the filter is controlled by a Low Frequency Oscillator (LFO). Different modes select different waveforms.



Click!



Filter Cutoff Depth

Controls the range of the filter – the amplitude of the LFO. On the far right the filter is off.

LFO Frequency

Controls frequency of the LFO. The LFO modulates the cut off frequency of a low pass filter.

Scroll >> to scroll different waveform combinations.

Long Click ...!
Return to main menu

Click ! Change sequence

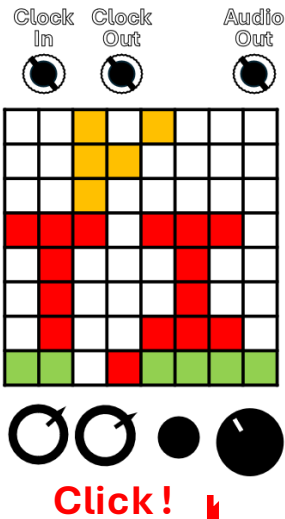
Scroll >> to hear different waveform combinations. Usually best to just hear what sounds best.

- Red – 4 x Sine waves
- Yellow – 4 x Triangle waves
- Green – 4 x Saw wave (This usually sounds the nicest)
- Blue – 4 x Square wave
- Cyan – 3 x Square wave 1 x White noise
- Pink – 3 x Saw wave 1x White noise
- Mauve – 4 x White noise

Menu Text: TI

Mode: Timing

This mode is used to control tempo. There are two different versions of the mode depending on if your Prioritron is acting as Conductor or a Player. The Conductor sets the tempo for all the connected Players. In conductor mode, scrolling will increase or decrease the tempo. In Player mode you can choose if your Prioritron players at double, half or quarter the tempo set by the conductor



In conductor mode

Filter Cutoff
Controls the frequency of the filter

Filter Envelope
Controls the amount of filter applied to the start of the note. Turn to far right to turn off filter.

Scroll >> to increase or decrease the tempo

Long Click ...! Edit start and end point of sequence

In player mode

Scroll >> to scroll through timing modes (double, normal, half and quarter speed)

Click! Select the timing mode (turns from Red to Green)

Long Click ...! Return to main menu

Long Click ...!

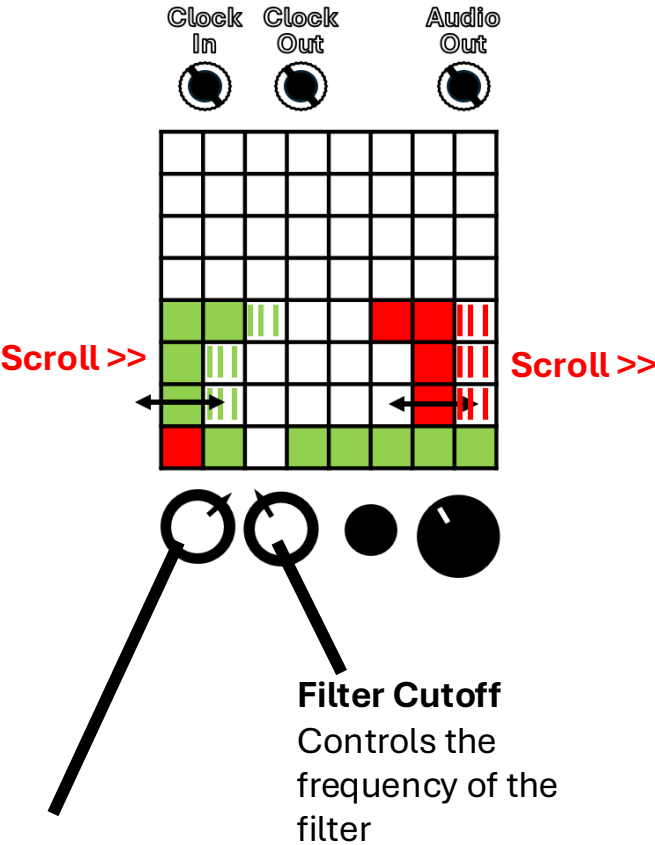
Click ! Change sequence

Filter Cutoff
Controls the frequency of the filter

Filter Envelope
Controls the amount of filter applied to the start of the note. Turn to far right to turn off filter.

Click ! Change sequence

This mode is reached through a long click in the timing mode. In this mode you are able to set the start and end points of the sequence, allowing you to make sequences that are shorter than 8 steps.



Scroll >> move, start, end or both.

Click! Toggle between moving the start, end or both

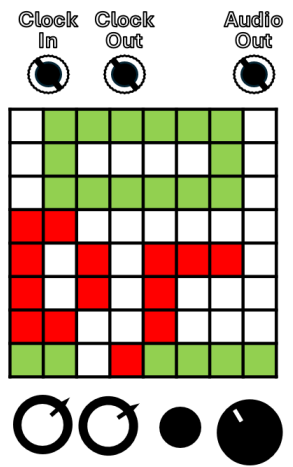
Long Click ...! Return to main menu

Filter Envelope
Controls the amount of filter applied to the start of the note. Turn to far right to turn off filter.

Menu Text: **Dr**

Mode: **Drums**

This is a special sequencer for the drums. The lower 5 drums are samples, click, clap, hi-hat, kick drum and snare. The top two drums use the synth oscillators to make a variety of weird crunches and zaps. In this mode the sequencer works a little differently as you can program any or all of the percussion to hit on any step. You can switch between 6 programmable drum sequences similar to when using the other synth modes.



Weird Drum 1 →

Weird Drum 2 →

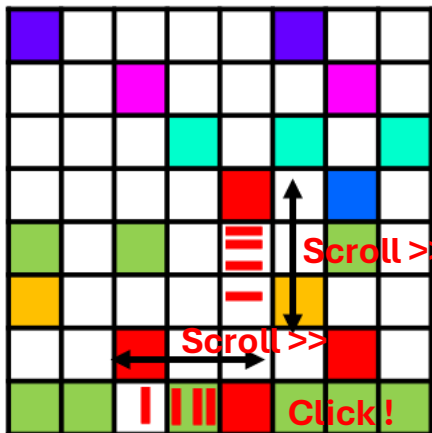
A Click →

Clap →

Hi-Hat →

Kick Drum →

Snare →



Pitch Bend – To the left this makes the pitch of Weird Drum 2 bend down, to the right it bends the pitch up.

Pitch – This sets the initial pitch of Weird Drum 1 and 2

Scroll >> moves the cursor between the steps on the bottom row or up and down between the instruments.

Click! Toggles the selected instrument on and off OR on the bottom row allows you to toggler between scrolling the steps and the instruments

Long Click ...! Return to main menu

Click ! Save Current Sequence and switch to next sequence (indicated by colour)

Long Click ...! Save Sequence

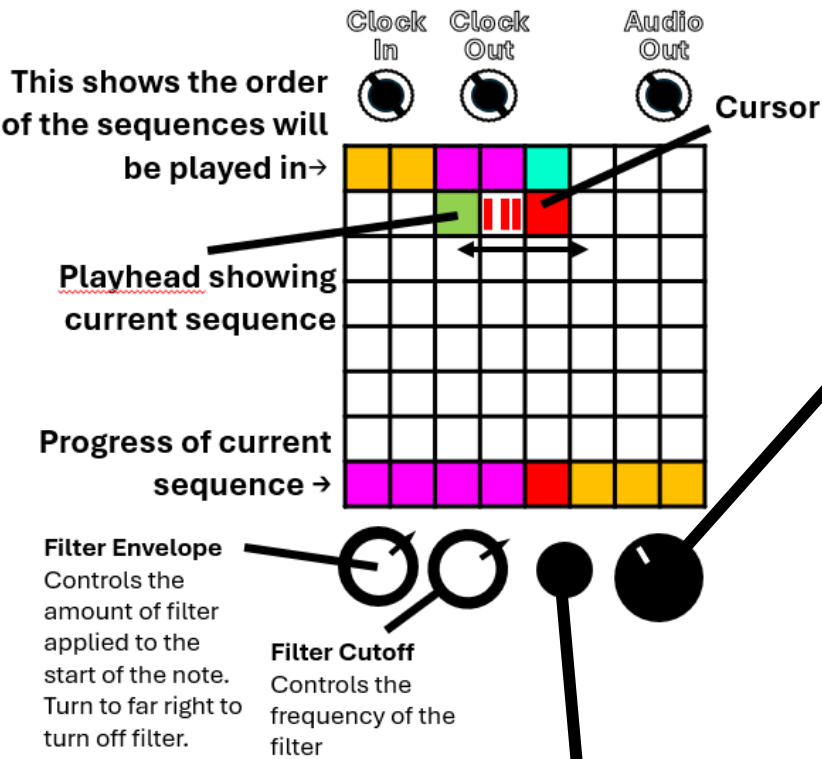
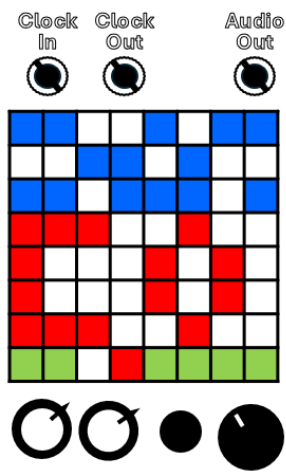
Hold and Scroll !>> If you hold this button and scroll then you will copy the current sequence into the sequence colour you are on when you release.

Menu Text: **Co**

Mode: **Controller**

This is a special sequencer mode that allows you to control the order that the coloured sequences that you have programmed will play.

This will work with drums or the synth sequences, depending on which one is currently playing. Pressing the button restarts the whole sequence from the start.



Click !

Scroll >> moves the cursor along the sequence and changing the programmed sequence.

Click! Toggles between changing the sequence and moving the cursor

Long Click ...! Return to main menu

Click ! Returns the play head to the start once the current sequence is complete.